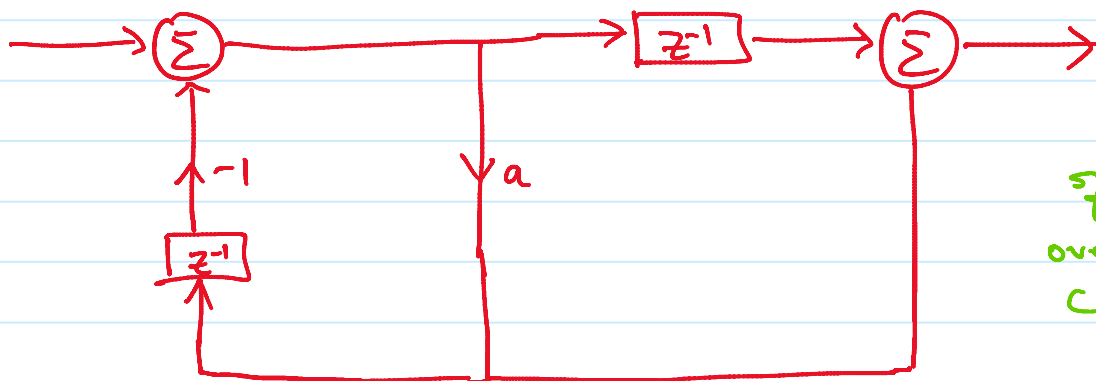


Tut 12
Filters

1. Given $\frac{a + z^{-1}}{1 + a z^{-1}}$ is realized in the form as given below



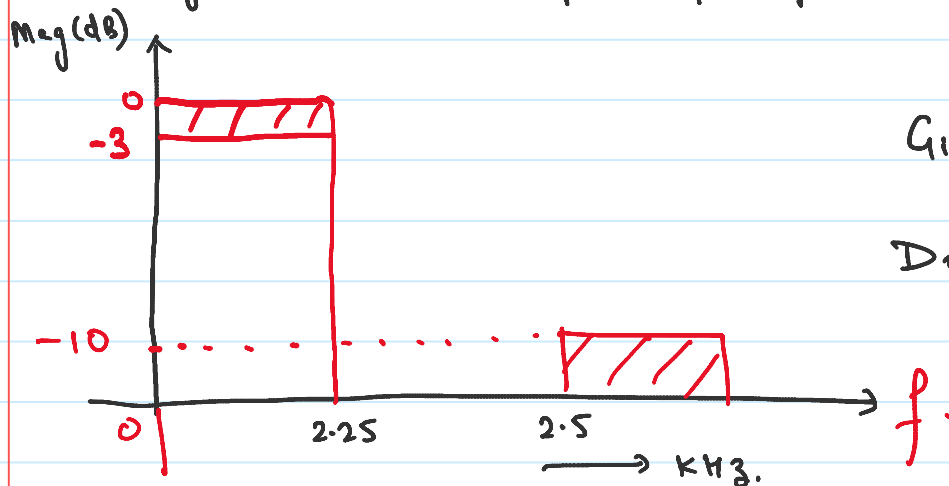
S.T. total no.
of delays in
overall structure
Can be reduced to
4.

Find Cascade realization of $A(z) = \left(\frac{a + z^{-1}}{1 + a z^{-1}} \cdot \frac{b + z^{-1}}{1 + b z^{-1}} \cdot \frac{c + z^{-1}}{1 + c z^{-1}} \right)$
by sharing the delays b/w adjacent all pass sections.

2. Transform single pole LP Butterworth filter with system function

$$H(s) = \frac{\Omega_p}{s + \Omega_p} \text{ into a BPF with upper } \Omega_u \text{ \& lower band edge } \Omega_l, \text{ respectively.}$$

3. A digital LPF has foll specifications



Given $f_s = 9 \text{ kHz}$.

Design Chebyshev digital LPF.